WORKED SOLUTION

STEP II 1998, Question 8

Pure • Vector Geometry

STEP II 1998, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q8

The key idea

Everything here is a matter of writing each line in the form "point + parameter × direction" and then choosing the parameter cleverly. The expressions are symmetric in **a**, **b**, **c**, so if one parameter value gives an answer that is symmetric in the three vectors, the same point lies on all three lines. That is exactly what "by inspection" is inviting us to look for.

Solution

The medians meet at R

The mid-point of BC has position vector $\frac{1}{2}(\mathbf{b}+\mathbf{c})$, so the line from A to it is

$$\mathbf{r} = \mathbf{a} + s\left(\tfrac{1}{2}(\mathbf{b}+\mathbf{c}) - \mathbf{a}\right) = (1-s)\mathbf{a} + \tfrac{s}{2}(\mathbf{b}+\mathbf{c}).$$

We want the three coefficients to be equal, so that the result is symmetric: $1 - s = \frac{s}{2}$ gives $s = \frac{2}{3}$, and the point is

$$\mathbf{r} = \tfrac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c}).$$

The lines from B and from C are obtained by permuting $\mathbf{a}, \mathbf{b}, \mathbf{c}$ cyclically, and the same choice $s = \frac{2}{3}$ gives the same symmetric point on each. So all three lines pass through the point R with position vector

$$\mathbf{r}_R = \tfrac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c}),$$

the centroid of the triangle. (Since A, B, C form a genuine triangle, no two of these lines coincide, so R is their single common point.)

The three parallel lines meet at Q

Let P have position vector $\mathbf{p}$. The three lines are

$$\begin{aligned}
&\text{through the mid-point of } BC, \text{ parallel to } PA: && \mathbf{r} = \tfrac{1}{2}(\mathbf{b}+\mathbf{c}) + \lambda(\mathbf{a}-\mathbf{p}),\\
&\text{through the mid-point of } CA, \text{ parallel to } PB: && \mathbf{r} = \tfrac{1}{2}(\mathbf{c}+\mathbf{a}) + \mu(\mathbf{b}-\mathbf{p}),\\
&\text{through the mid-point of } AB, \text{ parallel to } PC: && \mathbf{r} = \tfrac{1}{2}(\mathbf{a}+\mathbf{b}) + \nu(\mathbf{c}-\mathbf{p}).
\end{aligned}$$

Ask yourself What value of λ makes the first expression symmetric in $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$?

In the first line, $\mathbf{b}$ and $\mathbf{c}$ each carry the coefficient $\frac{1}{2}$ while $\mathbf{a}$ carries λ; matching them needs $\lambda = \frac{1}{2}$. Then

$$\mathbf{r} = \tfrac{1}{2}(\mathbf{a}+\mathbf{b}+\mathbf{c}-\mathbf{p}).$$

Taking $\mu = \frac{1}{2}$ and $\nu = \frac{1}{2}$ in the other two lines gives exactly the same vector. Hence all three lines pass through the point Q with

$$\mathbf{q} = \tfrac{1}{2}(\mathbf{a}+\mathbf{b}+\mathbf{c}-\mathbf{p}).$$

PQ meets the plane ABC at R

First, $Q \neq P$: if $\mathbf{q} = \mathbf{p}$ then $\mathbf{a}+\mathbf{b}+\mathbf{c} = 3\mathbf{p}$, so $\mathbf{p} = \mathbf{r}_R$ and P would be the point R, which lies in the plane ABC. So PQ is a genuine line, namely

$$\mathbf{r} = \mathbf{p} + t(\mathbf{q}-\mathbf{p}) = \mathbf{p} + t\left(\tfrac{1}{2}(\mathbf{a}+\mathbf{b}+\mathbf{c}) - \tfrac{3}{2}\mathbf{p}\right).$$

To kill the $\mathbf{p}$ terms we need $1 - \frac{3}{2}t = 0$, that is $t = \frac{2}{3}$, and then

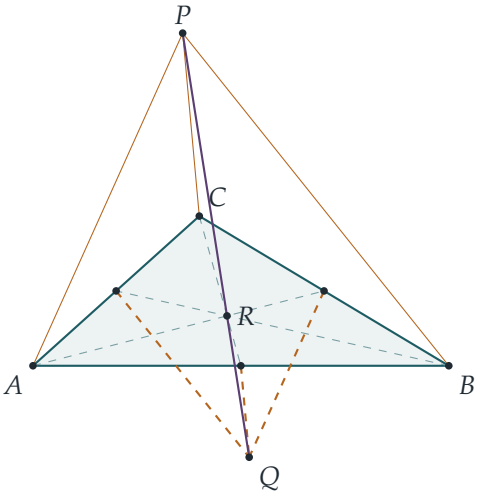
$$\mathbf{r} = \mathbf{p} + \tfrac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c}) - \mathbf{p} = \tfrac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c}) = \mathbf{r}_R.$$

So R lies on the line PQ. It also lies in the plane ABC (it is on the median from A, which lies in that plane).

Ask yourself Could PQ meet the plane somewhere else as well?

No. A line meets a plane in no points, exactly one point, or lies entirely in it. The line PQ does not lie in the plane ABC, because P is on it and P is not in the plane. So it meets the plane in at most one point, and that point is R. This proves that PQ meets the plane ABC at R.

As a bonus, $t = \frac{2}{3}$ tells us that R divides PQ in the ratio $PR : RQ = 2 : 1$; equivalently $\mathbf{q} - \mathbf{r}_R = -\frac{1}{2}(\mathbf{p} - \mathbf{r}_R)$.



Checks

- Each claimed intersection can be checked by substituting back: for instance, $\mathbf{q} - \frac{1}{2}(\mathbf{b}+\mathbf{c}) = \frac{1}{2}(\mathbf{a}-\mathbf{p})$, which is visibly parallel to PA.
- With $A=(0,0,0)$, $B=(3,0,0)$, $C=(0,3,0)$ and $P=(0,0,3)$ we get $R=(1,1,0)$ and $Q=(\frac{3}{2},\frac{3}{2},-\frac{3}{2})$; the point two-thirds of the way from P to Q is $(1,1,0)=R$, on the plane $z=0$.
- We also checked all the results with exact rational arithmetic on 2000 random choices of A, B, C, P: the line PQ always met the plane ABC at R, with parameter $t=\frac{2}{3}$.

Common slips

- Finding a point on two of the lines by solving equations and forgetting to check that it lies on the third.
- Showing that R lies on PQ but not explaining why it is *the* point where PQ meets the plane; this is where the condition that P is not in the plane is used.

Beyond STEP

There is a neat geometric reason behind the result. The enlargement with centre R and scale factor $-\frac{1}{2}$ (the map $\mathbf{x} \mapsto \mathbf{r}_R - \frac{1}{2}(\mathbf{x}-\mathbf{r}_R)$) sends A, B, C to the mid-points of BC, CA, AB, and sends any line to a parallel line. So it carries the lines PA, PB, PC, which meet at P, onto the three new lines, which must therefore meet at the image of P; that image is Q, and P, R, Q are collinear because the centre of an enlargement is collinear with every point and its image. Thinking of geometry through its transformations (affine maps and their fixed points) is a central idea of first-year linear algebra; Strang's

Introduction to Linear Algebra is a good place to see it developed.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.